

Demonstrating Virtual Streamer with Conversational and Tactile Interaction

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Figure 1: User Interactions. (a) a virtual streamer waving at the user, (b) a user doing a handshake, (c) high-five, and (d) a side hug.

ABSTRACT

This paper introduces an approach to an interactive virtual streamer that provides conversational and tactile interactions, demonstrating insights into levels of immersion and personalization for the user. User interaction is categorized into various types, such as the spatial nature of the virtual streamer's stream, along with conversational and tactile interactions with the streamer. We demonstrate the virtual streamer in VR and conclude the efficiency of user interactions with the virtual streamer via a scenario-based assessment.

Index Terms: Virtual Human, Haptic Interaction, Conversational Interaction, Avatar Interaction

1 INTRODUCTION

Streaming has evolved into a prevalent form of entertainment, offering users a more personal connection with content creators. Virtual streamers, or VTubers, are streamers who use an avatar instead of a face camera to animate their interactions, adding an engaging layer to the streaming experience. Traditionally, virtual streamers are controlled by humans where the avatar's actions, facial expressions, and lip movements are mapped to the human's motions and expressions. Recently, there has been a notable surge in virtual streamers powered by artificial intelligence (AI), which has gained massive popularity in the streaming world. These AI-powered virtual streamers operate autonomously with little to no human intervention to control the avatar's facial expressions, play games, and interact with users [3]. This is in contrast to traditional virtual streamers who rely on pre-set avatar mapping with their face and body, and require complete human intervention to stream games and interact with users.

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Although AI-powered virtual streamers are intelligent and can communicate with users while streaming games, they pose limitations in terms of user engagement and personalization. Since virtual streamers stream to a large audience, they cannot cater to each user's comments and, therefore, hinder effective user engagement. Furthermore, virtual streamers often face challenges in representing realism due to limitations in current technologies [1]. They operate behind a screen, which hinders users from directly and tactically interacting with virtual streamers, further decreasing user immersion.

We demonstrate incorporating an AI-powered virtual streamer into VR, allowing users to engage in visual, auditory, conversational, and tactile interactions, thereby adding another layer of realism to the stream. In our demo, a VR user will be immersed in VR where conversations are perceived through the user's speech and processed by a large language model (LLM); this model generates a response and returns it back to the user through speech, allowing the virtual streamer to communicate with the VR user. The virtual streamer is able to live stream games to the user using the Twitch platform. Moreover, the VR user can interact with the virtual streamer through touch using haptic gloves (bHaptics TactGlove DK1) and vests (bHaptics TactSuit X40), allowing for direct and effective user interaction, which fosters a dynamic streaming experience and user engagement (see Figure 1).

2 DESIGN

The goal of our approach is to provide an interactive virtual streamer in VR to increase immersion and user engagement. The virtual streamer is presented as an avatar to communicate with users through conversational and tactile interactions.

2.1 Virtual Environment

The virtual environment is created in Unity, featuring a room with a TV, desk, and chair (see Figure 2(a)). An interactive virtual streamer is presented as an avatar for conversational and full-body haptic interactions (see Figure 2(b)). A user with haptic gloves and a vest can enter the VR scene, watch a game stream on the TV, and interact with the avatar (see Figure 2(c)).

2.2 Streaming Service

The game is streamed on Twitch [2] on a server using OBS Studio. The Twitch server accesses the Twitch stream and projects it

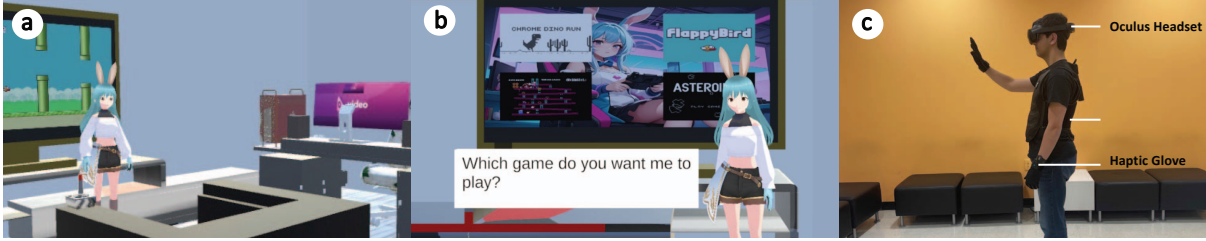


Figure 2: (a) A VR scene, (b) a virtual streamer is presented as an interactive avatar, (c) a user wearing haptic gloves and a vest.

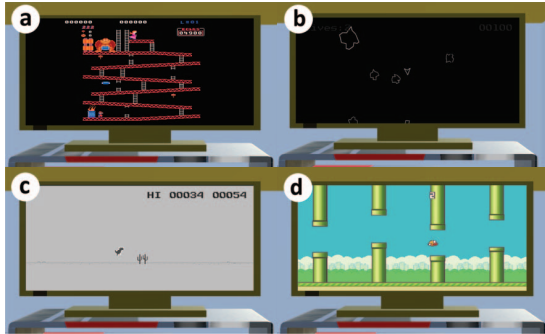


Figure 3: Four games: (a) Donkey Kong, (b) steroids, (c) Chrome Dino, and (d) Flappy Bird.

to the VR environment. We created four games: Donkey Kong, steroids, Chrome Dino, and Flappy Bird (see Figure 3). A user can watch these games on the TV via Twitch stream. The games can be switched using voice interaction. Each game has a machine learning (ML) model trained using ML Agents, allowing the streamer to play each game for the user's entertainment.

2.3 Conversation Module

Our virtual streamer engages in intelligent conversations with the user through a three-step process. It first converts the user's speech to text, then generates a response based on the text, and finally converts the response back to speech.

Speech to Text: The VR headset's microphone captures the user's speech, while a Python script in the background actively awaits and processes the audio input. The user's spoken words are transcribed into text using Python's 'SpeechRecognition' library.

Response Generation: The transcribed text is sent as input to a large language model (LLM). We used the GPT 3.5 LLM, which can hold conversations with the user in a manner comparable to human conversations.

Text to Speech: After generating a response, the virtual streamer replies to the user with speech. The speech was generated using a Neural2 voice using deep neural networks trained to generate realistic-sounding speech. The generated audio is then played through the VR headset's speakers. Using ULipSync, the virtual streamer's lips sync with the played audio.

2.4 Haptic Module

Our platform supports haptic interaction with the user. The avatar can detect when the user is within its proximity and collision. In the occurrence of a collision, the user will receive haptic feedback via a vest and gloves based on the type of collision.

Two types of haptic interaction are provided: a handshake and a hug. The avatar detects when the user's hand is close to the avatar's hand, prompting a vibration that simulates a handshake

through a glove. The glove uses actuators at the fingertips to produce these vibrations, which occur in a timed sequence. Likewise, the avatar detects when the user's headset is near the avatar's spine, thereby detecting the hug. The vest uses actuators in a sequence to simulate the feeling of having hands around the user's back, with the middle actuators vibrating first and the side actuators vibrating next. Subsequently, the gloves vibrate after the vest, mirroring the conclusive phase of the hug: the feeling of hands encircling the body. We used inverse kinematics to simulate a hug between the avatar and the user. This approach is to have the hands follow a target game object and use a hint that adjusts the forearm to be in a natural position, allowing more physical interaction in the VR environment.

In addition to the haptic interactions, the avatar waves when greeting the user. The streamer's avatar is rigged using VRoid Studio. We used the open-source library from Mixamo to apply three animations to the rigged avatar: idle, wave, and handshake. The avatar defaults to the idle position, but switches to the wave when greeting the user and to the handshake when it detects the user's hands near the streamer's hands. The avatar's eyes also blink occasionally as a feature.

3 DEMO EXPERIENCE

At the conference, we will demonstrate the proposed virtual environment with the stream. Attendees interested in the interactive demo will be briefed on the application and how they can interact with the virtual streamer. Then, they will be asked to put on the headset, the haptics gloves, and the haptics vest. The demonstrators will then set up the virtual environment and stream. The attendees can start interacting with the virtual streamer and watch the stream. They can interact with the streamer through dynamic conversation and tactile feedback. The demo will take at most 5 minutes per attendee, including setup and briefing.

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